



OWP | OFFICE OF  
WATER  
PREDICTION

---

**Data and Compute Architecture for Effective Use of  
Cloud Computing to Perform Benchmark Applications of  
the NextGen Framework**

*Nels Frazier, Mike Johnson, Justin Singh-Mohudpur, Keith Jennings,  
Austin Raney*



# Motivation

The **NextGen water resources modeling framework** provides extreme flexibility for modeling hydrology from local to continental scales.

The framework's ecosystem and tools have matured, and we need to start understanding how we apply them to formulate the National Water Model.

## Goal

Leverage public cloud compute resources for calibrating

**4 Hydrologic formulations** using

**100 iterations of DDS** search each over a

**5 year calibration** period plus a

**1 year validation** period in

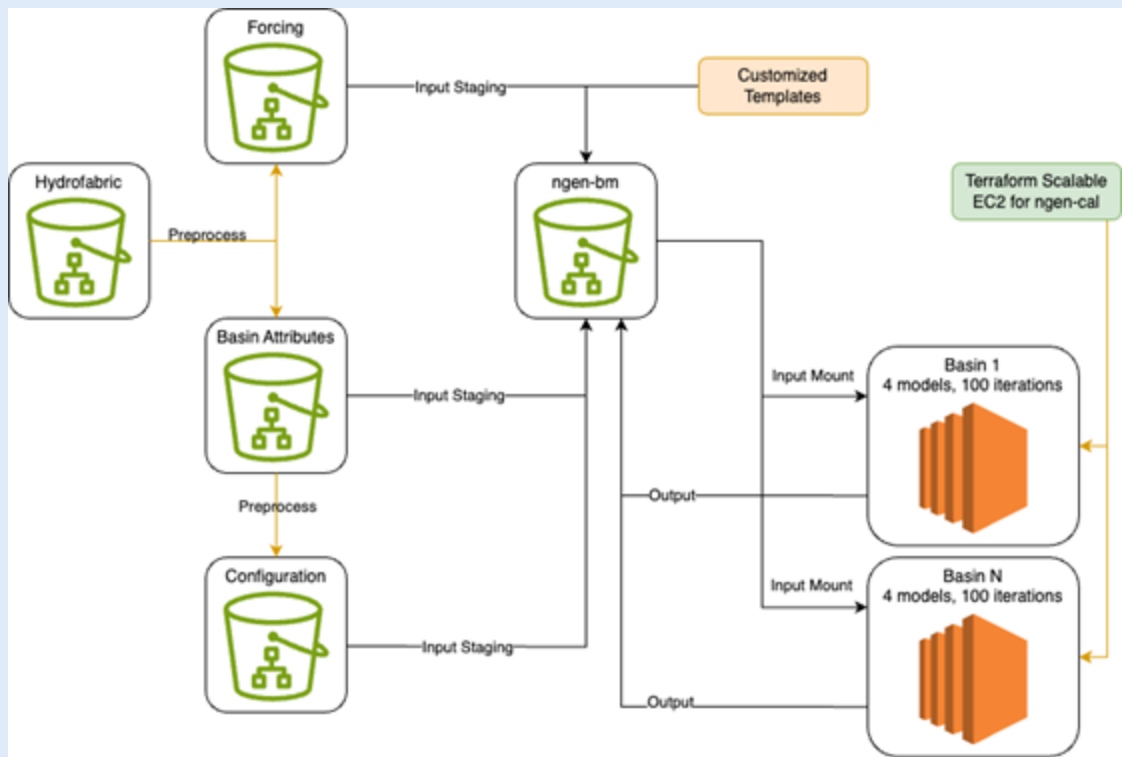
**31 Basins**

**3622 divides**



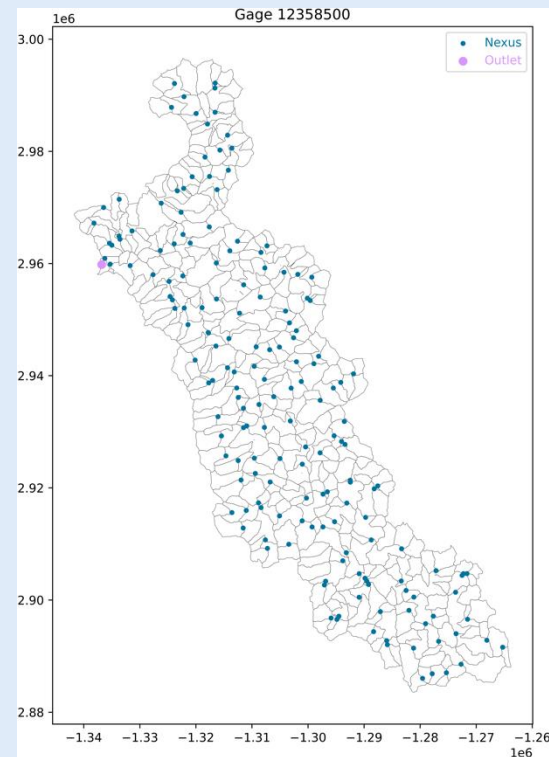
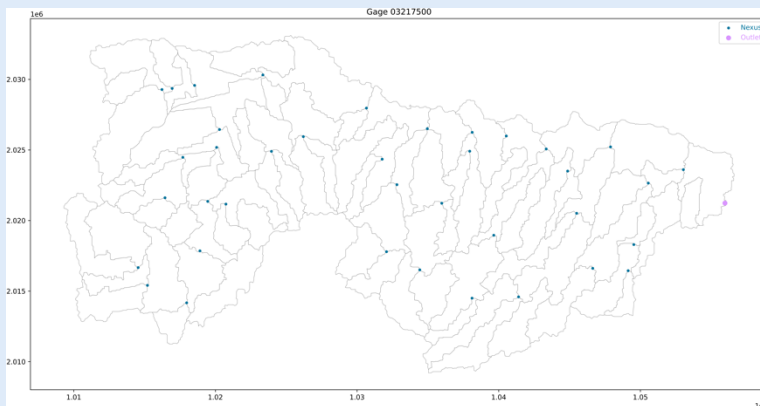
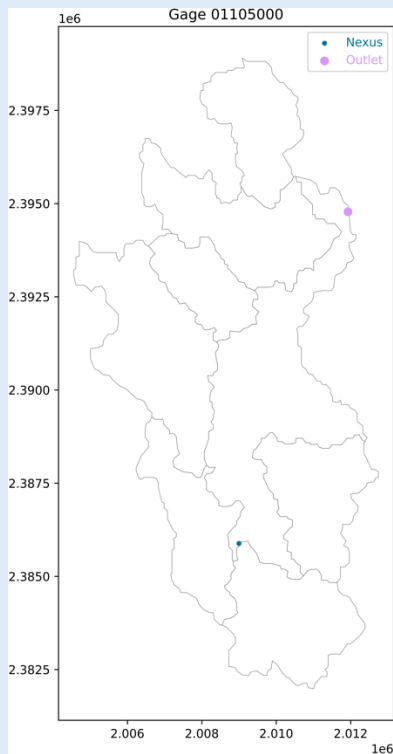
**OWP** | OFFICE OF  
WATER  
PREDICTION

# Design



- 1 AWS EC2 machine per basin
- Cloud storage
- Customized templates
- Fully automated deployment
- Pre-processing

# Areas of Interest



Processed 31 basins

|                 |                       |
|-----------------|-----------------------|
| Smallest (left) | ~74 km <sup>2</sup>   |
| Median (top)    | ~667 km <sup>2</sup>  |
| Largest (right) | ~2913 km <sup>2</sup> |



# Forcing (Pre Processing)

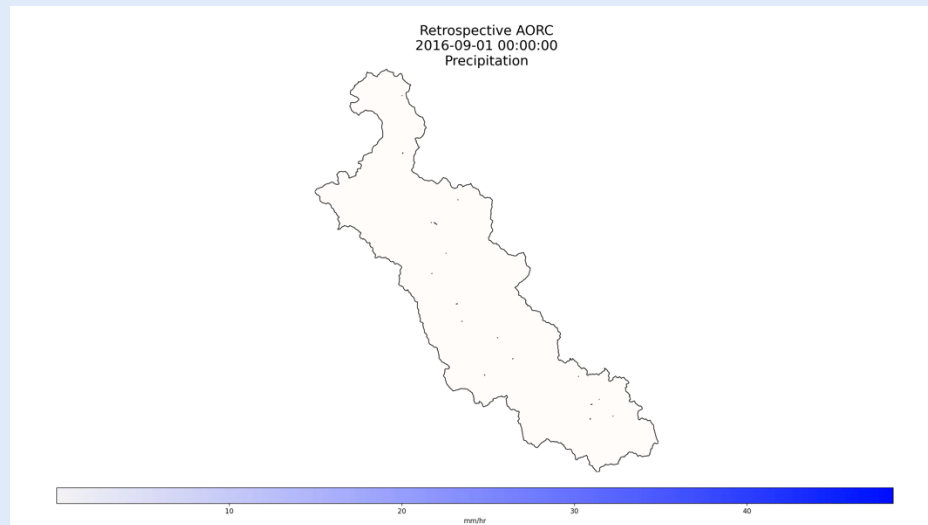
- AORC 1km forcing
- Zarr source data
- Exact Extract polygon rasterization
- Weighted averaging

Processed the **hourly forcings for the entire 40 year retrospective period** for each basin.

|                                            |            |
|--------------------------------------------|------------|
| Smallest Basin<br>(74.17 km <sup>2</sup> ) | 31 minutes |
|--------------------------------------------|------------|

|                                           |            |
|-------------------------------------------|------------|
| Median Basin<br>(666.81 km <sup>2</sup> ) | 25 minutes |
|-------------------------------------------|------------|

|                                             |                     |
|---------------------------------------------|---------------------|
| Largest Basin<br>(2913.72 km <sup>2</sup> ) | 2 hours, 30 minutes |
|---------------------------------------------|---------------------|





# Configuration, Templates, and Plugins

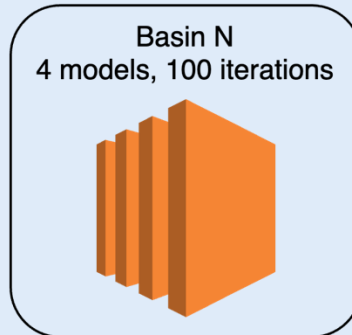
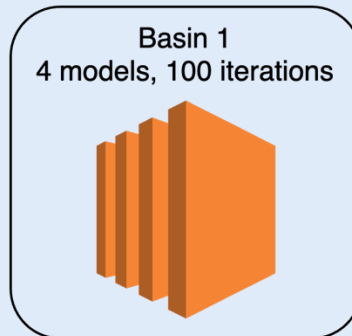
Final pre-processing stage using previous data staged on AWS S3

- Custom spatial processing for additional model parameters
- Programmatically **generate 18,110 BMI initialization files**

## Runtime Support Components

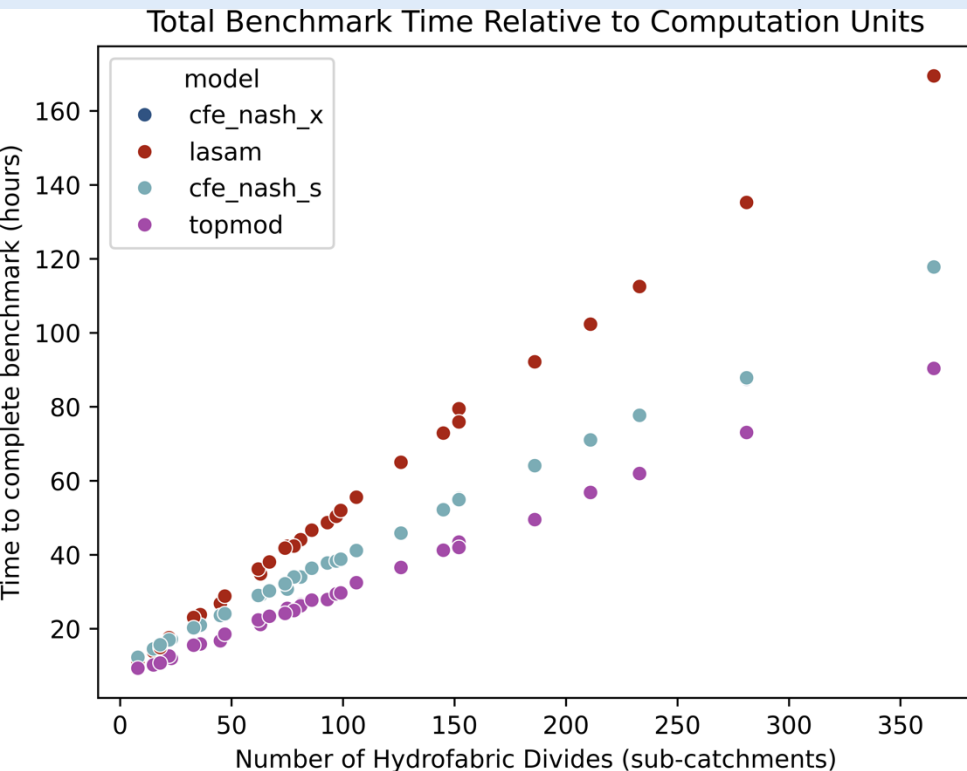
- Substitution templates for **easy, automatable reconfiguration**
- Custom calibration plugin for capturing **additional model performance metrics**
- Specialized cloud init functionality, configured with terraform

**"One click" deployment** of all benchmark simulations on AWS, start to finish!





# Results (Computational)



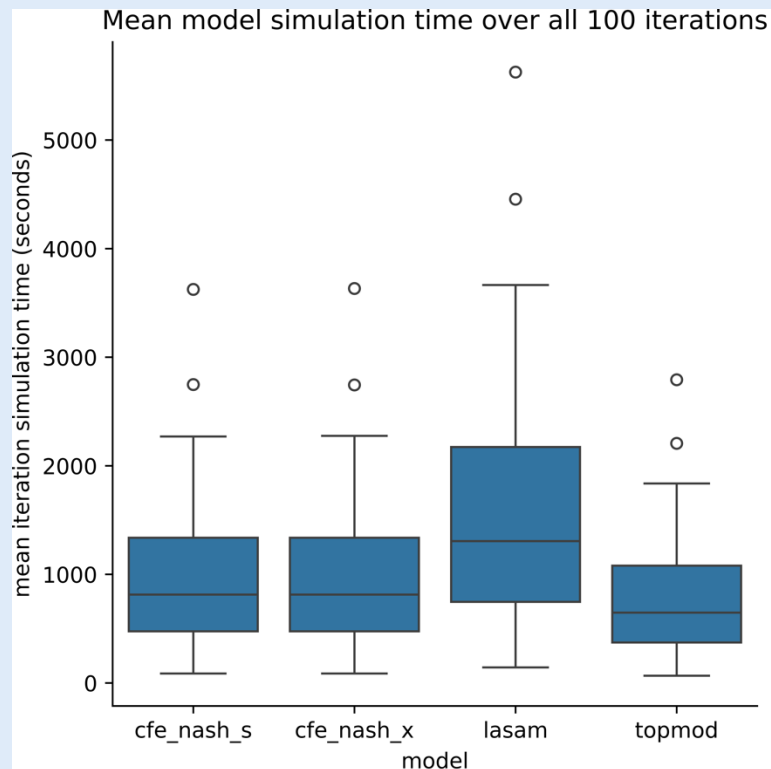
- On average, finished in **~41 hrs**
- Benchmark runtimes scale as basin size increases

|                |             |
|----------------|-------------|
| Smallest Basin | 8 divides   |
| Median Basin   | 81 divides  |
| Largest Basin  | 365 divides |

## Notes:

- The number of divides in a basin is strongly correlated with the total basin area
- cfe\_nash\_x and cfe\_nash\_s are very similar models, and they have nearly identical runtimes in this plot

# Results (Computational)



Additional output captured allows more detailed runtime analysis

- Average NextGen module simulation time of each model throughout the 100 calibration iterations for each basin.
- topmod fastest (marginally)
- lasam slowest
- Variation expected given the range of basin sizes simulated





# Cost

Estimated compute cost\*

- On demand EC2
- No wait time

**\$ 0.0001 per divide-year!**

\* over-estimate of minimal cost due to an initial bug that left some compute resources under-utilized, and required new compute resources to complete these runs

## New cost and usage report

Recent reports ▼

Save to report library

### Cost and usage graph [Info](#)

Total cost

**\$761.78**

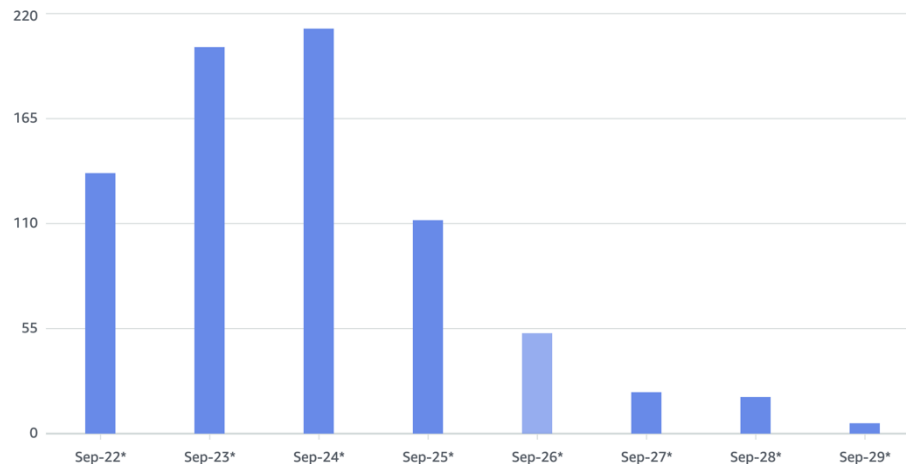
Average daily cost

**\$95.22**

Service count

**1**

Costs (\$)



## Cloud Bonus Features

- Tagged resource tracking
- Realtime analytics

# Success

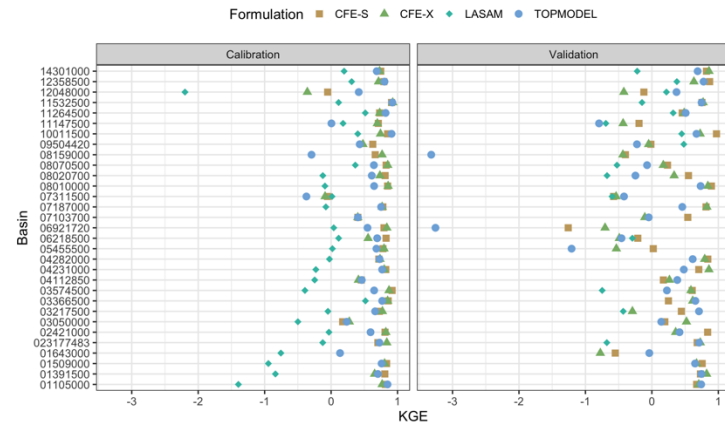
~ **2,042 years** *per basin* simulated

~ **63,306 total years** simulated

**104,345.84 km<sup>2</sup>** modeled over these times

**Reproducible** benchmark  
**scalable** to any gaged basin\*  
in the NextGen hydrofabric!

\*currently limited to headwater basins



## Improvements

Incorporate pre-processing steps into a complete end-to-end workflow.

Optimize EC2 machine types for cost/performance.



ngen-bm repo



OWP OFFICE OF  
WATER  
PREDICTION



OWP | OFFICE OF  
WATER  
PREDICTION



***Thank You!***

---



Nels Frazier



[nels.frazier@noaa.gov](mailto:nels.frazier@noaa.gov)



<https://water.noaa.gov>